

RUTHERGLEN, Australia

WMO: 958370

Lat: 36.1047S

Lon: 146.5094E

Elev: 175

StdP: 99.24

Time Zone: 10.00 (AUS)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	-2.2	-1.2	-4.1	2.7	12.0	-2.7	3.1	8.2	9.9	11.1	8.3	10.3	0.2	100	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	16.9	37.7	18.8	35.1	18.5	33.2	18.1	22.1	28.7	21.2	28.1	20.3	27.8	4.3	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	15.7	23.2	19.2	14.3	23.1	18.0	13.2	22.9	65.6	28.7	62.1	28.0	59.1	27.4	34.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.7	7.4	6.4	-3.9	41.1	1.3	1.9	-4.9	42.5	-5.7	43.6	-6.4	44.7	-7.4	46.1		
(5)			-4.0	24.0	1.4	2.8	-5.1	26.0	-5.9	27.6	-6.7	29.2	-7.7	31.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	23.4	22.6	19.3	14.7	10.6	8.2	7.6	8.2	10.3	13.2	17.6	20.4		
	DBStd	6.50	4.24	3.73	3.74	3.20	2.85	2.51	2.22	2.37	2.84	3.38	4.08	3.94		
	HDD10.0	279	0	0	0	2	27	67	81	67	28	7	0	0		
	HDD18.3	1844	7	6	33	115	241	305	334	315	241	164	62	22		
	CDD10.0	1967	416	352	288	144	45	11	6	11	38	105	228	322		
	CDD18.3	490	165	125	63	6	0	0	0	0	1	4	40	86		
	CDH23.3	7385	2417	1637	909	134	2	0	0	0	9	155	718	1403		
	CDH26.7	3502	1323	796	362	21	0	0	0	0	1	42	295	662		
(14) Wind	WSAvg	2.5	3.1	2.7	2.6	1.9	1.9	1.8	2.2	2.2	2.6	2.7	3.0	3.2		
(15) Precipitation	PrecAvg	578	43	30	32	42	61	48	67	61	56	57	44	46		
	PrecMax	956	192	148	100	160	163	145	179	122	132	153	107	145		
	PrecMin	262	0	0	0	4	4	4	6	7	3	0	0	0		
	PrecStd	176	42	33	26	34	43	33	42	31	30	35	30	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.3	39.6	35.8	29.2	22.5	17.9	16.2	19.5	24.5	31.4	35.3	38.9		
		MCWB	19.4	19.9	19.0	16.6	14.1	12.3	11.9	13.9	14.9	16.1	17.6	18.3		
	2%	DB	38.6	35.9	32.7	26.6	20.3	16.0	14.6	17.0	21.5	27.3	32.7	35.5		
		MCWB	19.1	19.3	18.5	16.0	14.0	11.8	11.3	12.5	14.8	15.4	17.0	17.9		
	5%	DB	35.8	33.6	30.5	24.7	18.5	14.7	13.5	15.3	19.3	24.6	30.2	32.8		
		MCWB	18.9	18.9	17.7	15.4	13.5	11.4	10.6	11.5	13.7	14.9	16.9	17.2		
	10%	DB	33.3	31.5	28.4	22.9	16.9	13.4	12.3	13.9	17.5	22.1	27.6	30.4		
		MCWB	18.3	18.4	16.9	14.9	12.6	10.8	9.9	10.7	12.9	14.4	16.2	16.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	24.1	22.1	19.2	16.7	14.3	13.7	14.7	17.2	19.5	21.4	22.4		
		MCDB	29.8	31.3	28.9	24.2	18.7	15.7	14.6	17.5	21.5	24.6	29.1	27.2		
	2%	WB	21.9	22.3	20.7	17.6	15.2	12.8	12.0	13.2	15.9	17.7	19.9	20.6		
		MCDB	29.6	28.5	26.5	22.7	18.3	14.7	13.6	15.9	19.6	22.8	26.3	26.9		
	5%	WB	20.9	21.3	19.6	16.7	14.1	12.0	11.0	12.1	14.6	16.6	18.7	19.4		
		MCDB	29.4	27.5	26.4	21.9	17.1	13.8	12.8	14.8	18.2	21.3	25.1	27.3		
	10%	WB	20.0	20.2	18.5	15.8	13.2	11.1	10.1	11.0	13.4	15.4	17.6	18.4		
		MCDB	29.6	27.2	25.6	21.4	16.2	13.1	11.8	13.4	16.8	20.5	24.1	27.5		
(35) Mean Daily Temperature Range	MDBR		16.9	15.6	15.7	15.1	12.7	10.9	9.9	11.4	13.5	15.4	16.3	16.8		
	5% DB	MCDBR	19.9	18.3	18.2	17.8	14.9	13.1	11.7	14.1	16.6	20.4	20.1	20.3		
		MCWBR	5.8	6.3	7.1	8.5	9.1	9.2	8.5	9.8	10.3	10.1	7.3	6.8		
	5% WB	MCDBR	14.2	12.6	14.0	14.4	11.9	10.6	9.9	12.1	14.0	14.7	14.4	14.6		
		MCWBR	5.4	5.6	6.6	7.6	7.8	7.9	7.5	8.8	9.6	9.0	6.5	6.5		
(40) Clear-Sky Solar Irradiance	taub		0.355	0.348	0.320	0.314	0.301	0.292	0.288	0.297	0.326	0.327	0.345	0.345		
	taud		2.449	2.493	2.567	2.544	2.554	2.584	2.596	2.550	2.453	2.476	2.467	2.475		
	Ebn at Noon		979	964	955	901	856	839	865	908	932	974	984	994		
	Edh at Noon		119	110	94	85	73	66	69	83	104	110	116	117		
(44) All-Sky Solar Radiation	RadAvg		7.68	6.73	5.60	4.03	2.77	2.11	2.28	3.19	4.55	6.01	6.96	7.71		
	RadStd		0.45	0.41	0.30	0.20	0.19	0.25	0.14	0.26	0.44	0.50	0.38	0.33		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+134	+71

Nomenclature: See separate page